

Arrays in Java

Arrays

An array is a collection of values with the same data type.

```
String[] animals = {"Giraffe", "Elephant",  
"Toucan"};
```

```
// Access an element via its index:  
System.out.println(animals[0]); // Prints:  
Giraffe
```

```
// Change an element value:  
animals[1] = "Lion";
```

```
// Find number of elements in an array:  
System.out.println(animals.length); //  
Prints: 3
```

```
// Traverse array using for loop:  
for (int i = 0; i < animals.length; i++) {  
    System.out.println(animals[i]);  
}  
/* Prints:  
Giraffe  
Lion  
Toucan  
*/
```

```
// Traverse array using for-each loop  
for (int i: animals) {  
    System.out.println(i);  
}  
/* Prints:  
Giraffe  
Lion  
Toucan  
*/
```

2D Arrays

A 2D array is an array that stores arrays of the same data type.

```
// Declare a 2D array that stores char arrays
char[][] letters = {{'A', 'a'}, {'B', 'x'},
{'C', 'c'}};
```

```
// Access an element via its index:
System.out.println(letters[0][1]); // Prints:
a
```

```
// Change an element by accessing its index:
letters[1][1] = "b";
```

```
// Iterate over 2D array in row-major order:
for (int i = 0; i < letters.length; i++){
    for (int j = 0; j < letters[0].length;
j++){
        System.out.print(letters[i][j]);
    }
}
// Prints: AaBbCc
```

```
// Iterate over 2D array in column-major
order:
for (int i = 0; i < letters[0].length; i++){
    for (int j = 0; j < letters.length; j++){
        System.out.print(letters[j][i]);
    }
}
// Prints: ABCabc
```